

Lie Groups and their relation to Lie Algebras

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1 Introduction

Lie groups were named after Norwegian mathematician Sophus Lie (1842-1899). Inspired by Galois theory, which demonstrated how discrete groups can simplify the solving of polynomial equations through symmetry and transformations of their roots. Lie extended this idea to differential equations using smooth manifolds and group theory.

2 What are Lie groups?

Definition: (Group)[1]

A group is a set S with a binary operation $\circ : S \times S \rightarrow S$ satisfying the following properties; identity, inverses, and associativity.

Definition: (Lie group)[2]

A Lie group is a smooth manifold whose underlying set of elements is equipped with the structure of a group such that the group multiplication and inverse-assigning functions are smooth functions.

Definition: (Homeomorphism)[3]

a homeomorphism $f : X \rightarrow Y$ is a continuous map of topological spaces such that there is an inverse $f^{-1} : Y \rightarrow X$ that is also a continuous map of topological spaces.

- Manifold [4]: A manifold is a topological space that is locally Euclidean (i.e., around every point, there is a neighbourhood that is topologically the same as the open unit ball in $\mathbb{R}^n$).
- A differentiable manifold [5] is a topological space which is locally homeomorphic to a Euclidean space and such that the gluing functions which relate these Euclidean local charts to each other are differentiable functions, for a fixed degree of differentiability. If one considers arbitrary differentiability, then one speaks of smooth manifolds.

Examples of smooth manifolds: torus, sphere, Klein bottle, Mobius strip

Examples of topological spaces that aren't manifolds: $\mathbb{Q}$, figure-8 on $\mathbb{R}^2$, cone

2.1 Elements of a Lie group

The elements of Lie groups are usually transformations (e.g. rotations) that are represented by matrices, e.g. the Lie group $SO(3)$ is a 3-dimensional smooth manifold and its elements can be used to rotate vectors in $\mathbb{R}^3$ while preserving their lengths.

Consider the following matrix:

$$R_z(\theta) = \begin{pmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

This matrix rotates vectors anticlockwise about θ in the xy plane, and preserves the z coordinate. This matrix has determinant 1 and is orthogonal. Hence this matrix is in the Lie group.[6]

3 Relation to Lie algebras

Lie algebras relate to Lie groups by providing an easier way to study their local structure through tangent spaces, exponential maps, and isomorphisms.

3.1 Tangent space

Informally, the Tangent space at a point of a differentiable manifold is a vector space containing the possible directions that vectors can pass tangentially through that point, it is commonly defined through other methods like tangents curves and derivations.[7]

3.2 Exponential map

$\exp: \mathfrak{g} \rightarrow G$ $\exp(X) = \gamma(1)$ with $\gamma: \mathbb{R} \rightarrow G$ the unique continuous subgroup homomorphism whose tangent vector at the identity is equal to X .

3.3 Isomorphisms

$SU(2)$ and $SO(3)$ are not isomorphic as Lie groups yet have isomorphic Lie algebras:

$$\mathfrak{su}(2) = \text{span}_{\mathbb{R}} \left\{ E = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, H = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, F = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} \right\}$$

Where $[H, E] = 2F$, $[E, F] = 2H$ and $[F, H] = 2E$

$$\mathfrak{so}(3) = \text{span}_{\mathbb{R}} \left\{ \eta = \begin{pmatrix} 0 & \sqrt{2} & 0 \\ -\sqrt{2} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \zeta = \begin{pmatrix} 0 & 0 & \sqrt{2} \\ 0 & 0 & 0 \\ -\sqrt{2} & 0 & 0 \end{pmatrix}, \delta = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & -2 & 0 \end{pmatrix} \right\}$$

Where $[\zeta, \eta] = 2\delta$, $[\eta, \delta] = 2\zeta$ and $[\delta, \zeta] = 2\eta$

The linear map $\phi: \mathfrak{su}(2) \rightarrow \mathfrak{so}(3)$ with $\phi(H) = \zeta$, $\phi(E) = \eta$, $\phi(F) = \delta$ is a Lie algebra isomorphism.

4 Applications

4.1 In mathematics

- Fundamental for the study of Representation Theory
- In topology, Lie groups provide a way of classifying manifolds.
- Fluid dynamics, important in the study of turbulent flows and other complex ideas
- In differential geometry, Lie groups are important for understanding how geometric objects can be transformed through symmetry.

4.2 In other fields

- Particle physics and Gauge theories, describing fundamental forces in physics (i.e. $U(1)$ describes electromagnetism, and $SU(2)$ and $SU(3)$ relate to weak and strong nuclear forces)
- $SO(3)$ is fundamental in modelling the rotations in three dimensions, including the symmetries of atomic orbitals, angular momentum and quantum mechanics.
- Robotics, for motion planning, particularly $SO(3)$ and $SE(3)$ can be used, for example, to plan the motion of robotic arms and ensuring their smooth and effective movement
- Computer science and graphics, used to model the rotations of animations ensuring realistic movement.

References

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